

Bounded-Support Hahn Arithmetic

Cofinal descent, optimal support thresholds,
and maximal transcendence

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Abstract

Let G be a nonzero set-sized ordered abelian group, let K have characteristic zero, and let $\mathcal{B}_G(K)$ be the ring of Hahn series whose support is bounded above in G . Put $\mathcal{F}_G(K) = \text{Frac } \mathcal{B}_G(K)$ and $\kappa = \text{cf}(G)$. We construct 2^κ elements of $K((t^G))$, all with positive integer coefficients and one common support of order type κ , that are algebraically independent over $\mathcal{F}_G(K)$. No smaller support cardinality can support even one transcendental element over this base. Every valuation ball contains such an independent family. In particular, the transcendence degree is the continuum for real-exponent Hahn fields over $\mathbb{R}$ or $\mathbb{C}$.

A complementary descent theorem proves that, whenever $H \subseteq G$ is cofinal and $K \subseteq L$, the fields $K((t^H))$ and $\mathcal{F}_G(L)$ are linearly disjoint over $\mathcal{F}_H(K)$. Polynomial relation ideals and finite algebraic degrees therefore survive simultaneous enlargement of coefficients and cofinal exponent groups. We also classify when $\mathcal{B}_G(K)$ is a field, reduce its unit theory to the highest Archimedean quotient, and prove a conditional GCD-descent theorem relevant to a published pre-Schreier question. Concrete Conway-normal-form specializations give independent families of actual surreal numbers, including maximal families of size 2^κ at every infinite regular cardinal.

Status. Complete arguments for the stated new results are given relative to explicitly identified standard Hahn-field and Conway-normal-form foundations. The proposed novelty is not a certified priority claim. This draft is not refereed or Lean-verified. The GCD application is explicitly conditional and is not used in any transcendence theorem.

In this collection. A single-source report, built from one manuscript; every label carries the prefix `bst:`. Its relation to the neighbouring reports is Section 9; its provenance is Section 1.4.

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1 The problem, the results, and their provenance

A Hahn series can contain infinitely many terms while its exponents remain in one bounded interval. Such a series is not a polynomial, and adjoining all such series produces a much larger base than a rational function field. The question pursued here is what remains genuinely transcendental after *every* bounded-support series has been admitted as a coefficient, and how that answer changes when the ambient exponent group changes.

Write

$$\mathcal{B}_G(K) = \{f \in K((t^G)) : \text{supp}(f) \text{ is bounded above in } G\}, \quad \mathcal{F}_G(K) = \text{Frac } \mathcal{B}_G(K). \quad (1)$$

The phrase “bounded support” always refers to the order on the specified exponent group. It does not mean finite support, bounded coefficients, or a bound on the cardinality of the support. In particular it is not the cardinal restriction sometimes called a λ -bounded Hahn field.

1.1 The principal theorems

Here is the theorem package in a form that can be read before its proofs. All fields and groups in these statements are sets. Cardinal assertions are made in set theory with choice.

Theorem 1.1 (Optimal-support independence; proved in Section 5). *Let K have characteristic zero and $G \neq 0$ be an ordered abelian group. Set $\kappa = \text{cf}(G)$. There are a strictly increasing cofinal sequence $(a_\alpha)_{\alpha < \kappa}$ of positive elements of G and positive integers $c_A(\alpha)$, for $A \subseteq \kappa$, such that*

$$Y_A = \sum_{\alpha < \kappa} c_A(\alpha) t^{a_\alpha} \quad (A \subseteq \kappa)$$

are algebraically independent over $\mathcal{F}_G(K)$. Their common support has order type κ . The minimum possible cardinality, and the minimum possible order type, of the support of a series transcendental over $\mathcal{F}_G(K)$ are both κ . Every nonempty valuation ball contains a family of 2^κ elements algebraically independent over $\mathcal{F}_G(K)$.

The strength is not a cardinality comparison between the base and the larger field: they can have the same cardinality. It is the construction of a free polynomial algebra over the entire bounded-support fraction field, with integer-coefficient witnesses and an optimal support requirement.

Theorem 1.2 (Cofinal coefficient–exponent descent; proved in Section 3). *Let $K \subseteq L$ be fields and let $0 \neq H \subseteq G$ be ordered abelian groups. If H is cofinal in G , multiplication induces an injection*

$$\mathcal{F}_G(L) \otimes_{\mathcal{F}_H(K)} K((t^H)) \longrightarrow L((t^G)).$$

Consequently

$$\mathcal{F}_G(L) \cap K((t^H)) = \mathcal{F}_H(K).$$

For every finite tuple from $K((t^H))$, its polynomial relation ideal over $\mathcal{F}_G(L)$ is the ideal generated by its relations over $\mathcal{F}_H(K)$. Finite algebraic degrees are preserved. In characteristic zero the cofinality hypothesis is also necessary for the displayed multiplication map to be injective.

For example, with $H = \mathbb{Z}$, $G = \mathbb{R}$, $K = \mathbb{Q}$, and $L = \mathbb{C}$, every algebraic relation among ordinary rational-coefficient Laurent series over $\mathcal{F}_{\mathbb{R}}(\mathbb{C})$ already comes from relations over $\mathbb{Q}(t)$. This is a simultaneous coefficient and exponent extension statement, not merely an intersection statement.

The structural companion is an exact distinction between the existence and absence of a highest scale. The ring $\mathcal{B}_G(K)$ is a field precisely when G has no order unit. In that case it is the union of the Hahn fields over proper convex subgroups. For divisible G and real closed K , that union is real closed; for algebraically closed K of characteristic zero, it is algebraically closed. When an order unit exists, the units are controlled by the highest Archimedean quotient; see Section 7.

1.2 What is already known, and what is proposed here

L’Innocente and Mantova study bounded-support localization and its arithmetic in [1]. Their Proposition 2.4.5 already gives a cofinality obstruction to representing all Hahn series by fractions of one-sided series. That nonrationality obstruction is *not* claimed here as new. The proposed strengthening is the $2^{\text{cf}(G)}$ -sized algebraically independent family with one optimal common support, together with the coefficient–exponent descent and its exact converse. Their Proposition 3.5.1 supplies the upper-support multiplicativity used only in the unit and GCD discussion.

The repository audit used the documentation catalogue and the closest transcendence report at revision

048b72cf7cbfc8ab246e4f73788c10460cb3f6e0.

At that revision the catalogue listed 36 reports. Raw placement in commit 30dfb4f added this report and three others, bringing that historical count to 40; their editorial assembly followed in commit ed88b8f. Its tail-span report [5] constructs differential independence using independent square classes in coefficient extensions. Here the coefficients of the witnesses can all be positive integers, the base is a bounded-support fraction field, and the mechanism is support separation. These are different base fields and different assertions; neither report should be described as automatically subsuming the other. The catalogue also contains entire-function and cofinality results [6], but a ring of entire functions is not the ring in (1). Section 9 states precisely which results of these reports overlap with those here and which differ.

The proposed priority claims are deliberately limited. Targeted searches did not identify the combined descent theorem or the sharp independent-family theorem in these formulations. This is not an exhaustive MathSciNet, Zentralblatt, thesis, or citation-network audit. The elementary coding lemma, standard linear-disjointness algebra, Hahn field construction, real/algebraic closedness criteria, and Conway normal form are background, not new discoveries. A later repository revision, 6e34651fe640, was observed during the audit; its recorded change concerned Hahn–Hilbert spectral theory. No claim of a full source-by-source audit of that later revision is made.

1.3 Dependencies and the boundary of the claim

The main independence proof uses only Hahn multiplication, finite polynomial algebra, cardinal cofinality, and a fully proved finite-pattern coding lemma. It does not use a GCD theorem, the upper-support multiplicativity result, any surreal derivation, or analytic convergence. The descent proof is coefficientwise and works in arbitrary characteristic. Standard Hahn-field closedness and Conway normal form are invoked only where named. Appendix A proves an implication toward a published GCD/pre-Schreier problem; it does not supply the missing real-exponent GCD input.

1.4 Provenance of this report

This report is built from one manuscript, *Bounded-Support Hahn Arithmetic: cofinal descent, optimal support thresholds, and maximal transcendence*, a research draft prepared with ChatGPT and dated 22 September 2026. It arrived as item 07 of the nineteenth batch of external manuscripts. Commit 30dfb4f placed the raw source and artifacts; commit ed88b8f assembled the article and added its first repository PDF. Its own repository audit is pinned to revision 048b72cf7cbfc8ab246e4f73788c10460cb3f6e0 (Appendix B). The manuscript contributed every theorem, proof, example, appendix and non-claim printed here; nothing was merged from another source, so no result is printed twice and there was no choice between competing proofs.

The editorial changes made when the report entered the collection are: every label received the prefix `bst`; the actual surreal witnesses of Section 8, written ξ_A and ξ_p in the manuscript, are written η_A and η_p here, because the tail-span report already uses ξ_A for different numbers (Section 9.1); the terminology conventions of Section 2.3 and the comparison of Section 9 were added; and statements about the repository that had become stale were corrected, keeping the original pin as provenance. The delivered programs and records are kept byte-identical in code/ and data/; Section 10 gives the file names. No theorem statement was strengthened or weakened at assembly. The subsequent proof review expands the arguments and corrects the convergence and comparison wording. It also replaces the former 52c7ab6 placement citation, which is unavailable in the local history, by the distinct placement and assembly commits above.

2 Hahn series and bounded-support algebra

2.1 Conventions

An ordered abelian group is written additively. A Hahn series over K is a formal expression

$$f = \sum_{g \in G} f_g t^g$$

whose support $\{g : f_g \neq 0\}$ is well ordered. Addition is coefficientwise and multiplication is the Hahn convolution. At each exponent the relevant sum is finite. Finite unions and finite sumsets of well-ordered supports are well ordered. These facts give its ring operations. For a nonzero series, factor out its leading monomial to write $f = at^g(1 + h)$ with $a \in K^\times$ and h of strictly positive support (or $h = 0$). The Hahn–Neumann positive-support lemma makes $\sum_{n \geq 0} (-h)^n$ strongly summable and permits coefficientwise telescoping, giving the inverse $a^{-1}t^{-g} \sum_{n \geq 0} (-h)^n$. This standard inverse construction completes the field $K((t^G))$; see [1, Section 2.1]. It uses no valuation convergence of the finite geometric sums. We use the conventional valuation

$$v(f) = \min \text{supp}(f) \quad (f \neq 0), \quad v(0) = +\infty.$$

Thus $v(fg) = v(f) + v(g)$ and $v(f + g) \geq \min(v(f), v(g))$. The convention at zero is stated explicitly because some treatments of one-sided series use a different special-symbol convention.

A family of Hahn series is *strongly summable* when its combined support is well ordered and only finitely many summands contribute at any exponent. A sum along a strictly increasing sequence of exponents is strongly summable, independently of whether its finite partial sums converge in a chosen topology. All transfinite sums below are Hahn sums in this sense.

For $f \in K((t^G))$ and $\delta \in G$, write

$$f_{\leq \delta} = \sum_{g \leq \delta} f_g t^g.$$

A nonzero Hahn support has a least element. Therefore “bounded above” in (1) is equivalent to containment in some order interval $[\ell, u] \subseteq G$.

Lemma 2.1 (Elementary support bounds). *The set $\mathcal{B}_G(K)$ is a subring of $K((t^G))$, contains K and every monomial, and is closed under taking arbitrary subseries. If $\text{supp}(f) \subseteq [\ell, u]$ and $\text{supp}(g) \subseteq [\ell', u']$, then*

$$\text{supp}(fg) \subseteq [\ell + \ell', u + u'].$$

Every finite family of bounded-support series has common lower and upper support bounds.

Proof. Every exponent of a product is a sum of one exponent from each factor. Cancellation only removes terms and cannot create an exponent outside the indicated interval. A finite union is bounded by the minimum of its lower bounds and the maximum of its upper bounds. Negation leaves the support unchanged. Every subset of a well-ordered set is well ordered, proving the subseries assertion. These observations also cover zero by using arbitrary bounds for its empty support. $\square$

Proposition 2.2 (Localization and density). *For any nonzero G ,*

$$\mathcal{B}_G(K) = t^G K((t^{G^{\leq 0}})).$$

Both $\mathcal{B}_G(K)$ and $\mathcal{F}_G(K)$ are dense in $K((t^G))$ for the valuation topology.

Proof. If $\text{supp}(f) \leq u$, then $t^{-u}f$ has nonpositive support. Conversely a monomial translate of a nonpositive support is bounded above. For density, $f_{\leq \delta} \in \mathcal{B}_G(K)$ and either $f = f_{\leq \delta}$ or $v(f - f_{\leq \delta}) > \delta$. Hence every basic valuation ball contains a bounded-support element. The assertion for the fraction field follows. Here $t^G K((t^{G^{\leq 0}}))$ denotes the set of monomial translates of the nonpositive-support subring. It is its localization at the monomials t^{-g} for $g \geq 0$: choose the upper bound $u \geq 0$ above, so $f = (t^{-u}f)/t^{-u}$. This specifies the localization used in Appendix A. $\square$

There is no assertion here that bounded-support series are finite sums. For example

$$\sum_{n \geq 2} t^{1-1/n} \in \mathcal{B}_{\mathbb{Q}}(\mathbb{R})$$

has infinitely many exponents in $[1/2, 1)$. Nor is density an algebraicity statement. The main theorem will show that an enormous transcendental extension can remain after adjoining this dense subring's entire fraction field.

2.2 Cofinality

Definition 2.3. For a nonzero ordered abelian group G , its cofinality $\text{cf}(G)$ is the least cardinality of a subset unbounded above in G . An ordered subgroup H is *cofinal* if for every $g \in G$ some $h \in H$ satisfies $h > g$. An *order unit* is an element $u > 0$ such that for every $g \in G$ some positive integer n satisfies $|g| \leq nu$.

Lemma 2.4. *The cardinal $\kappa = \text{cf}(G)$ is infinite and regular. There is a strictly increasing positive cofinal sequence $(g_\alpha)_{\alpha < \kappa}$. Every subset of G of cardinality less than κ is bounded above. An ordered subgroup is either cofinal or bounded above.*

Proof. No finite subset is cofinal, since it has a maximum and G has positive elements. The last two assertions follow from the definitions. Start with a cofinal family of size κ and recursively choose a value exceeding its next member and all previously chosen values. At stage $\alpha < \kappa$ the latter set has size less than κ , so a bound exists. This gives the increasing sequence. If κ were singular, a cofinal subsequence of its indices of cardinality $\text{cf}(\kappa) < \kappa$ would give a smaller cofinal subset of G , a contradiction. $\square$

For a nonzero subgroup of $\mathbb{R}$, every positive element is an order unit and the cofinality is $\aleph_0$. The converse implication from countable cofinality to the existence of an order unit is false: a countable ascending union of strictly increasing Archimedean scales has no highest scale. Section 8 gives concrete examples inside **No**.

2.3 Terminology shared with other reports

The following conventions fix one meaning for each word and symbol that is used differently elsewhere in the collection.

1. *Cofinality*. $\text{cf}(G)$ is the least cardinality of a subset of the ordered set G that is unbounded above. This is exactly the invariant $\text{cf}(\Gamma)$ of the entire-functions report [6], whose definition uses “every element is at most some member”; for a nonzero ordered group, which has no greatest element, the two definitions agree.
2. *Cofinal*. The adjective is used for a subset or sequence of G that is unbounded above, and for an ordered subgroup $H \subseteq G$ with the same property. It is used once for a set of *indices*: in Lemma 5.1 a set of ordinals is cofinal in the initial ordinal κ . The entire-functions report expressly excludes that ordinal-index meaning from its own word (its remark `ent:rem:one-name`); here it is confined to that lemma and the proofs that cite it.
3. *Order unit*. The definition above agrees with `ent:def:order-unit`: $|g| \leq nu$ for all g is equivalent to $\{\mu : \mu \geq 1\}$ being unbounded above. An order unit forces $\text{cf}(G) = \aleph_0$, and the converse fails, in both reports.
4. *Cofinite* is not *cofinal*. The tail-span report’s cofinite span is an intersection over finite deletions of coefficient indices; nothing in this report deletes finitely many indices.
5. *Bounded*. “Bounded support” always means bounded above in the named exponent group, as in (1), and never bounded above in all of **No** (Section 8). It is not a bound on coefficients or on valuations, and not the cardinal restriction of a λ -bounded Hahn field.
6. *Symbols*. $\mathcal{B}_G(K)$ and $\mathcal{F}_G(K)$ always carry both the exponent group and the coefficient field. The collection’s notation guide [8] writes F_Γ for the full real Hahn field $\mathbb{R}((t^\Gamma))$, and the tail-span report writes B and B_0 for full Hahn fields $M((t^\Gamma))$ and $\mathbb{Q}((t^\mathbb{Q}))$. Neither is meant by the calligraphic letters here: $\mathcal{F}_G(K)$ is the fraction field of the bounded-support ring, a proper subfield of $K((t^G))$ in characteristic zero.

3 Cofinal coefficient–exponent descent

The following result is useful independently of the lacunary constructions. It explains why a bounded-support coefficient cannot conceal a new relation among series supported on a cofinal subgroup.

3.1 Coset projections and coefficient functionals

Let $K \subseteq L$ be fields, $H \subseteq G$ an ordered subgroup, and $r \in G$. Define the shifted coset projection

$$\pi_r(f) = \sum_{h \in H} f_{r+h} t^h \in L((t^H)). \quad (2)$$

It is defined for every $f \in L((t^G))$: a subset of its support remains well ordered, and translation preserves that property. This is not asserted to be a ring homomorphism: for $H = \mathbb{Z} \subset G = \mathbb{Q}$, one has $\pi_0(t^{1/2}) = 0$ but $\pi_0((t^{1/2})^2) = t$. Its useful property is the module identity below.

Lemma 3.1 (Projection identity). *For $z \in L((t^H))$ and $b \in L((t^G))$,*

$$\pi_r(bz) = \pi_r(b)z.$$

If H is cofinal and $b \in \mathcal{B}_G(L)$, then $\pi_r(b) \in \mathcal{B}_H(L)$.

Proof. A term of bz at an exponent in $r + H$ can only use a term of b in that same coset, because all exponents of z lie in H . At every exponent the convolution is finite, so selecting the coset commutes with it. If the support of b is bounded above by $u \in G$, then the shifted support is bounded above in G by $u - r$. Cofinality supplies an H -valued bound. $\square$

Let $\lambda : L \rightarrow K$ be any K -linear functional. Apply it to each coefficient to obtain

$$\lambda_* : L((t^H)) \longrightarrow K((t^H)).$$

This map can kill support terms but introduces none. In particular it preserves boundedness of support. For $z \in K((t^H))$ it satisfies

$$\lambda_*(bz) = \lambda_*(b)z. \quad (3)$$

The reason is again the finiteness of each convolution coefficient, now combined with K -linearity. No multiplicativity of λ is needed. For any nonzero $a \in L$, a K -linear functional with $\lambda(a) \neq 0$ exists: extend a to a basis and prescribe its value to be 1.

3.2 The descent theorem

Theorem 3.2 (Simultaneous descent). *Let $K \subseteq L$ be fields and let $0 \neq H \subseteq G$ be cofinal. Every finite family of elements of $K((t^H))$ linearly independent over $\mathcal{F}_H(K)$ remains linearly independent over $\mathcal{F}_G(L)$. Equivalently, the natural multiplication map*

$$\mathcal{F}_G(L) \otimes_{\mathcal{F}_H(K)} K((t^H)) \longrightarrow L((t^G)) \quad (4)$$

is injective.

Proof. Let $z_1, \dots, z_m \in K((t^H))$ be linearly independent over $\mathcal{F}_H(K)$. Suppose there is a nonzero linear relation over $\mathcal{F}_G(L)$. After multiplying by a common nonzero denominator, it becomes

$$\sum_{j=1}^m b_j z_j = 0, \quad b_j \in \mathcal{B}_G(L), \quad (5)$$

with some $b_{j_0} \neq 0$. Choose an exponent $r \in \text{supp}(b_{j_0})$ and set $a = (b_{j_0})_r \neq 0$. Applying π_r to (5) gives

$$\sum_j \pi_r(b_j) z_j = 0,$$

where all projected coefficients belong to $\mathcal{B}_H(L)$. Choose a K -linear functional $\lambda : L \rightarrow K$ with $\lambda(a) = 1$. Applying λ_* and using (3) yields

$$\sum_j \lambda_*(\pi_r(b_j)) z_j = 0.$$

Its coefficients belong to $\mathcal{B}_H(K) \subseteq \mathcal{F}_H(K)$. The coefficient of t^0 in $\lambda_*(\pi_r(b_{j_0}))$ is 1, so the relation is nonzero. This contradicts the assumed independence.

To verify the tensor formulation directly, write a tensor using a finite $\mathcal{F}_H(K)$ -linearly independent family of second factors. If its image is zero, the first part of the proof forces every first-factor coefficient to vanish. The converse equivalence follows by the same representation. This agrees with the standard definition of linear disjointness [3]. $\square$

Corollary 3.3 (Intersection, relations, and algebraic degree). *Under the hypotheses of Theorem 3.2:*

1. $\mathcal{F}_G(L) \cap K((t^H)) = \mathcal{F}_H(K)$ inside $L((t^G))$.
2. For a finite tuple z in $K((t^H))$, let $I_K(z)$ and $I_L(z)$ be its relation ideals over $\mathcal{F}_H(K)$ and $\mathcal{F}_G(L)$, respectively. Then

$$I_L(z) = I_K(z) \mathcal{F}_G(L)[X_1, \dots, X_m].$$

3. Algebraic independence of any family in $K((t^H))$ is preserved in both directions by this base enlargement.
4. If $z \in K((t^H))$ is algebraic of degree d over $\mathcal{F}_H(K)$, it has degree d over $\mathcal{F}_G(L)$, with the same monic minimal polynomial.

Proof. For the intersection statement, $z \notin \mathcal{F}_H(K)$ makes $1, z$ linearly independent over $\mathcal{F}_H(K)$, hence over $\mathcal{F}_G(L)$; it cannot then belong to $\mathcal{F}_G(L)$. The reverse inclusion is immediate.

Let $E = \mathcal{F}_H(K)$ and $F = \mathcal{F}_G(L)$. The quotient $E[X]/I_K(z)$ embeds in $K((t^H))$. Tensoring that inclusion with the E -vector space F preserves injectivity. By Theorem 3.2, the resulting algebra still embeds in $L((t^G))$. Its presentation is $F[X]/I_K(z)F[X]$, so its evaluation kernel is exactly the asserted ideal. This proves the second assertion and then the third. Finally $1, z, \dots, z^{d-1}$ remain linearly independent, while the old degree- d equation still holds, proving the last assertion. $\square$

For a finite Galois subextension of $K((t^H))/\mathcal{F}_H(K)$, the same argument preserves the degree and Galois group after compositum with $\mathcal{F}_G(L)$. Indeed, the tensor product embeds in the Hahn field by descent, so it is a domain. It is finite-dimensional over $\mathcal{F}_G(L)$ of the original degree, hence a field: multiplication by a nonzero element is an injective linear map and therefore surjective. Its distinct base-field automorphisms extend by the identity on $\mathcal{F}_G(L)$, and their number already equals that degree. This observation does not assert that every algebraic germ extension is Galois.

3.3 The ordinary Laurent-series case

Corollary 3.4 (One cofinal scale). *If $u > 0$ is an order unit of G and $K \subseteq L$, then $K((t^u))$ and $\mathcal{F}_G(L)$ are linearly disjoint over $K(t^u)$. In particular*

$$K((t^u)) \cap \mathcal{F}_G(L) = K(t^u).$$

Proof. The subgroup $\mathbb{Z}u$ is cofinal. Its bounded well-ordered subsets are finite, so $\mathcal{B}_{\mathbb{Z}u}(K) = K[t^u, t^{-u}]$ and its fraction field is $K(t^u)$. Apply Theorem 3.2. $\square$

Example 3.5 (An algebraic germ does not become a bounded fraction). Take $s = t^u$ and characteristic zero. The binomial series

$$z = (1 + s)^{1/2} = 1 + \frac{1}{2}s - \frac{1}{8}s^2 + \frac{1}{16}s^3 - \dots \in \mathbb{Q}[[s]]$$

has degree two over $\mathcal{F}_G(L)$ for every characteristic-zero extension $\mathbb{Q} \subseteq L$ and every G in which u is an order unit. To check its degree over $\mathbb{Q}(s)$, observe that $1 + s$ is not a rational square: in a coprime quotient of polynomials, the multiplicity of the factor $s + 1$ would be even on a square and odd on $1 + s$. Corollary 3.3 then preserves degree two. Thus it is algebraic but not itself in the bounded-support fraction field. In contrast, $(1 - s)^{-1}$ is in that fraction field, although its support is unbounded.

3.4 Why noncofinal enlargement behaves oppositely

If H is not cofinal, the entire group H is bounded above in G . Consequently

$$K((t^H)) \subseteq \mathcal{B}_G(L) \subseteq \mathcal{F}_G(L). \quad (6)$$

Every old Hahn series becomes a bounded-support coefficient in the enlarged workspace. Once Theorem 5.3 has established $\mathcal{F}_H(K) \neq K((t^H))$ in characteristic zero, this proves the converse in Theorem 1.2: choose z in that difference. The pair $1, z$ is independent over the smaller base but dependent over the larger one. Explicitly, $1 \otimes z - z \otimes 1$ is a nonzero tensor sent to zero by multiplication. To see that it is nonzero, extend $1, z$ to a basis of $K((t^H))$ over $\mathcal{F}_H(K)$ and use the linear functional taking 1 to zero and z to one; applying it to the second tensor factor gives 1. The exceptional subgroup $H = 0$ is excluded because its Hahn field and fraction field are both just K .

This dichotomy is about the actual embedding of exponent groups. Merely saying that both groups are divisible or both have countable cofinality does not imply that the embedding is cofinal.

4 The support-separation mechanism

This section supplies the estimate that upgrades the earlier nonrationality obstruction to algebraic independence. It is entirely algebraic. Even when a truncation has infinitely many terms, its support has the bound used in the estimate.

4.1 Separated exponent systems

Definition 4.1. Let κ be an infinite cardinal. A *separated exponent system* in G consists of positive elements a_α, b_α for $\alpha < \kappa$ such that:

1. a_α is strictly increasing and cofinal, and $a_\beta \leq b_\alpha$ whenever $\beta < \alpha$;
2. for every positive integer d and every $w \in G$, eventually in α one has

$$a_\alpha > db_\alpha + w, \quad a_{\alpha+1} > da_\alpha + w. \quad (7)$$

Every infinite cardinal is a limit ordinal, so $\alpha + 1 < \kappa$ for all $\alpha < \kappa$.

Lemma 4.2 (Existence at the exact cofinality). *Every nonzero ordered abelian group admits a separated exponent system of length $\kappa = \text{cf}(G)$.*

Proof. Fix a strictly increasing positive cofinal sequence g_α from Lemma 2.4.

First suppose $\kappa = \aleph_0$, and index by $n \geq 0$. Choose $b_n > 0$ strictly exceeding g_n and, when $n > 0$, a_{n-1} . Set $a_n = (n+2)b_n$. These elements are increasing and cofinal, and earlier a_j are at most b_n . Given d, w , eventually $b_n > w$ and $n+2 \geq d+1$, so $a_n - db_n \geq b_n > w$. At the next index the same estimate gives $a_{n+1} > db_{n+1} + w \geq da_n + w$. Both inequalities hold eventually.

Now suppose κ is uncountable. At stage α , the earlier a_β , together with g_α , form a set of size less than κ . Choose a positive strict upper bound b_α for them. The countable set $\{nb_\alpha : n \geq 1\}$ also has size less than κ , so choose a_α strictly above every one of its elements. This gives a strictly increasing cofinal sequence. For fixed d, w , eventually $b_\alpha > w$; then $a_\alpha > (d+1)b_\alpha > db_\alpha + w$. The estimate at $\alpha + 1$, and $b_{\alpha+1} > a_\alpha$, give the second inequality of (7). $\square$

Example 4.3 (Factorial spacing). If u is an order unit, the concrete choices

$$a_n = n!u, \quad b_n = (n-1)!u \quad (n \geq 2)$$

form a separated exponent system. For fixed d , $a_n - db_n = (n-d)(n-1)!u$ eventually exceeds every $w \in G$. The next-step inequality is checked in the same way.

4.2 A polynomial cannot bridge two separated support bands

Fix a separated exponent system $(a_\alpha, b_\alpha)_{\alpha < \kappa}$. Let

$$P(X) = \sum_v p_v X^v \in \mathcal{B}_G(K)[X_1, \dots, X_m]$$

be nonzero of total degree $d \geq 1$. Its top homogeneous part is denoted P_d . Choose $\ell, u \in G$ such that every p_v is supported in $[\ell, u]$. For coefficient vectors $c(\alpha) = (c_1(\alpha), \dots, c_m(\alpha)) \in K^m$, put

$$Y_j = \sum_{\alpha < \kappa} c_j(\alpha) t^{a_\alpha}.$$

Zero entries cause no problem: each support is a subset of the well-ordered set of a_α .

Lemma 4.4 (Support-band detection). *Suppose an index α satisfies*

$$a_\alpha > db_\alpha + (u - \ell), \quad a_{\alpha+1} > da_\alpha + (u - \ell), \quad (8)$$

and $P_d(c(\alpha)) \neq 0$ in $K((t^G))$. Then $P(Y_1, \dots, Y_m) \neq 0$.

Proof. Define the earlier part and the truncation through α by

$$A_j = \sum_{\beta < \alpha} c_j(\beta) t^{a_\beta}, \quad T_j = A_j + c_j(\alpha) t^{a_\alpha}.$$

Every A_j is supported in $[0, b_\alpha]$ and every T_j in $[0, a_\alpha]$. Expanding in an auxiliary *polynomial* variable Z , write

$$P(A_1 + c_1(\alpha)Z, \dots, A_m + c_m(\alpha)Z) = \sum_{k=0}^d Q_k Z^k. \quad (9)$$

The coefficient of Z^d is exactly $P_d(c(\alpha))$, so $Q_d \neq 0$. In a monomial of degree $|v|$, a contribution to Q_k uses k copies of Z and $|v| - k$ factors from the A_j . Reaching degree d in Z therefore uses only top-degree monomials and no A_j factors. Every coefficient Q_k is a finite sum of products, each consisting of a p_v , a scalar, and at most d factors chosen from the A_j . Lemma 2.1 therefore gives

$$\text{supp}(Q_k) \subseteq [\ell, u + db_\alpha] \quad (0 \leq k \leq d).$$

After substituting $Z = t^{a_\alpha}$, these supports lie in the bands

$$[\ell + ka_\alpha, u + db_\alpha + ka_\alpha]. \quad (10)$$

The first inequality in (8) says that successive bands are strictly disjoint. Since $Q_d \neq 0$, choose $g \in \text{supp } Q_d$. The nonzero coefficient at $g + da_\alpha$ lies in the last band and receives no contribution from any earlier band. Hence $P(T_1, \dots, T_m) \neq 0$.

A direct bound using the original polynomial and the supports of T_j shows that

$$\text{supp}(P(T)) \subseteq [\ell, u + da_\alpha]. \quad (11)$$

Meanwhile every tail $Y_j - T_j$ is either zero or has valuation at least $a_{\alpha+1}$. Expand each difference $Y^\nu - T^\nu$ as a sum of products containing at least one such tail. All other series factors have nonnegative valuation. Thus

$$v(P(Y) - P(T)) \geq \ell + a_{\alpha+1}. \quad (12)$$

By the second inequality in (8), this lower bound is strictly above the upper bound in (11). The nonzero bounded-support series $P(T)$ cannot be canceled by the tail. Hence $P(Y) \neq 0$. $\square$

Remark 4.5 (What the proof does not assume). Neither $P(T)$ nor the earlier parts A_j need have finite support. No supremum of their supports in G is required; any displayed upper bound suffices. There is no rearrangement of a non-summable family and no appeal to a numerical limit. All polynomial expansions are finite, and each coefficientwise product is a Hahn convolution.

Corollary 4.6 (A one-series version). *Let (a_n) be positive and strictly increasing, and suppose that for every $d \geq 1$ and $w \in G$, eventually $a_{n+1} > da_n + w$. If every $c_n \in K$ is nonzero, then $Y = \sum_n c_n t^{a_n}$ is transcendental over $\mathcal{F}_G(K)$.*

Proof. For a putative relation clear denominators to obtain a nonzero polynomial $P \in \mathcal{B}_G(K)[X]$. The case of constant P is impossible. The same tail estimate as (12) implies $P(T_n) = 0$ for every sufficiently large n , since otherwise its bounded support could not be canceled. Here T_n is the finite truncation through n , supported at most at a_n . These truncations are pairwise distinct: the difference of two of them has the nonzero coefficient c_j at the first newly included exponent. A nonzero polynomial over the field $K((t^G))$ has only finitely many roots. This contradiction proves the claim. $\square$

The following comparison was added in batch 31. The independent-copies report [11] prints the same tail estimate and root count at every cofinality (its `isc:thm:cofinalgap`) and claims no priority for it. Let K be an arbitrary field, finite fields included, let $G \neq 0$ be set-sized, and let $(a_\alpha)_{\alpha < \text{cf}(G)}$ be a strictly increasing cofinal family in $G^{>0}$ such that, for every $d \geq 1$ and $w \in G$, eventually $a_{\alpha+1} > da_\alpha + w$. Such a family exists (its `isc:lem:gapscale`). Then $1 + \sum_{\alpha < \text{cf}(G)} t^{a_\alpha}$, with all coefficients 1, is transcendental over $\mathcal{F}_G(K)$. Corollary 4.6 also works over any field and allows arbitrary nonzero coefficients, but it is indexed by $\mathbb{N}$ and its hypothesis makes (a_n) cofinal, so it needs $\text{cf}(G) = \aleph_0$. Theorem 5.3 covers every cofinality but needs characteristic zero. The single-series statement at uncountable cofinality over a field of positive characteristic, in particular a finite field, is therefore not stated in this report. It does not touch the multivariable problem, and finite coefficient fields remain outside the independence theorem (Section 10).

The multivariable argument needs more than this root-count observation. One must ensure that the top homogeneous polynomial does not vanish on all chosen coefficient vectors. The next section supplies exactly that certificate, without any algebraic independence assumption on the coefficients themselves.

5 Independent coefficient codes and the sharp theorem

5.1 A finite-pattern coding lemma

The following elementary construction is an independent-family coding argument. Its role is to realize any prescribed finite tuple of positive integers at cofinally many positions. It is not being claimed as a new set-theoretic principle.

Lemma 5.1 (Cofinal finite-pattern realization). *For any infinite cardinal κ , there are functions*

$$c_A : \kappa \longrightarrow \mathbb{N}_{>0} \quad (A \subseteq \kappa)$$

such that for every finite list of distinct subsets $A_1, \dots, A_m$ and every $(q_1, \dots, q_m) \in \mathbb{N}_{>0}^m$, the set

$$\{\alpha < \kappa : c_{A_j}(\alpha) = q_j \text{ for all } j\}$$

has cardinality κ , and in particular is cofinal in κ .

Proof. Let D_κ consist of triples (s, h, ρ) where s is a finite subset of κ , $h : \mathcal{P}(s) \rightarrow \mathbb{N}_{>0}$ is a table, and $\rho < \kappa$. There are κ finite subsets of κ , countably many tables for each finite s , and κ choices for ρ . Hence $|D_\kappa| = \kappa$. Fix a bijection

$$\alpha \longmapsto (s_\alpha, h_\alpha, \rho_\alpha) \quad \text{from } \kappa \text{ to } D_\kappa$$

and set

$$c_A(\alpha) = h_\alpha(A \cap s_\alpha). \quad (13)$$

For each pair $A_j \neq A_k$, choose a point in their symmetric difference. The finite set s of these points makes $A_j \cap s$ pairwise distinct. A table h can therefore take the prescribed value q_j at each of them; set its remaining values to 1. Every triple (s, h, ρ) realizes the target tuple, independently of ρ . There are κ such triples, which proves the cardinality assertion. A bounded subset of the initial ordinal κ has cardinality smaller than κ , so the realizing indices are cofinal. Regularity of κ is not needed for this coding lemma; cardinality κ already rules out a bounded subset of its initial ordinal. For a single prescribed set, the separating set s may be empty. Distinct sets give distinct coefficient streams: prescribe values 1 and 2 at one common realizing index. $\square$

Lemma 5.2 (Integer grids detect polynomials). *Let E be a field of characteristic zero. A nonzero polynomial $Q \in E[X_1, \dots, X_m]$ is nonzero at some point of $\mathbb{N}_{>0}^m$. More precisely, if each variable has degree at most d , any product of m sets of $d + 1$ distinct elements of E contains a nonzero value.*

Proof. For no variables, the assertion concerns a nonzero constant and is immediate. For one variable, a nonzero polynomial of degree at most d has at most d roots. For the induction step, regard Q as a polynomial in its last variable. If it vanishes at every point of the product grid, then for each fixed tuple of the first $m - 1$ variables all its coefficients vanish, by the one-variable statement. The induction hypothesis applied to those coefficient polynomials makes each identically zero, a contradiction. The positive integers are distinct in a characteristic-zero field. $\square$

5.2 Independence over all bounded-support fractions

Theorem 5.3 (The sharp support theorem). *Let K have characteristic zero and let $0 \neq G$ be an ordered abelian group. Put $\kappa = \text{cf}(G)$. Choose a separated exponent system as in Lemma 4.2 and a coefficient code as in Lemma 5.1. Then*

$$Y_A = \sum_{\alpha < \kappa} c_A(\alpha) t^{a_\alpha} \quad (A \subseteq \kappa) \quad (14)$$

are algebraically independent over $\mathcal{F}_G(K)$. Consequently

$$\text{trdeg}_{\mathcal{F}_G(K)} K((t^G)) \geq 2^\kappa. \quad (15)$$

Every member has positive integer coefficients and exactly the common support $\{a_\alpha : \alpha < \kappa\}$, of order type κ .

Proof. First, each Y_A is a valid Hahn series: the strictly increasing map $\alpha \mapsto a_\alpha$ identifies its image with the well-ordered initial ordinal κ , and characteristic zero makes every prescribed positive integer coefficient nonzero. Thus its support is exactly that image. Fix distinct $A_1, \dots, A_m$ and a nonzero polynomial over $\mathcal{F}_G(K)$. Clear a common nonzero denominator to obtain a nonzero polynomial $P \in \mathcal{B}_G(K)[X_1, \dots, X_m]$. If its total degree is zero, its evaluation is already nonzero. Otherwise let its degree be $d \geq 1$ and choose common coefficient support bounds $[\ell, u]$.

Its top homogeneous part P_d is a nonzero polynomial over the field $K((t^G))$. By Lemma 5.2, there is a tuple of positive integers $q = (q_1, \dots, q_m)$ such that $P_d(q) \neq 0$. Lemma 5.1 realizes this exact tuple at cofinally many indices α . The gap inequalities (8) hold at every sufficiently large index. Choose a realizing index after that threshold. Lemma 4.4 then gives

$$P(Y_{A_1}, \dots, Y_{A_m}) \neq 0.$$

This handles every finite subfamily and every nonzero polynomial, which is algebraic independence. There are 2^κ subsets A , proving the lower bound. Finally the coefficients are positive integers, hence nonzero in K , so none of the designated exponents disappears. $\square$

For the remaining corollaries in this section, retain the hypotheses $\text{char } K = 0$, $G \neq 0$ and $\kappa = \text{cf}(G)$, and the family Y_A of Theorem 5.3.

Corollary 5.4 (Optimal cardinality and order type). *Among all $z \in K((t^G))$ transcendental over $\mathcal{F}_G(K)$, the least support cardinality is $\text{cf}(G)$ and the least support order type is the initial ordinal $\text{cf}(G)$.*

Proof. A support of cardinality less than $\kappa = \text{cf}(G)$ is bounded above by definition of cofinality, so its series already belongs to $\mathcal{B}_G(K)$. Theorem 5.3 supplies a transcendental series with support of cardinality and order type exactly κ . An ordinal smaller than the initial ordinal κ has smaller cardinality, proving the order-type assertion as well. $\square$

Thus countable supports suffice exactly when the value group has countable cofinality. When the cofinality is uncountable, every countable-support series is already a coefficient in the base, regardless of how sparse its support looks inside a smaller subgroup.

Corollary 5.5 (Arbitrary bounded perturbations). *For any choices $b_A \in \mathcal{F}_G(K)$ and $u_A \in \mathcal{F}_G(K)^\times$, the family $(u_A Y_A + b_A)_{A \subseteq \kappa}$ is algebraically independent over $\mathcal{F}_G(K)$. In particular arbitrary bounded-support changes, or deletion of an arbitrary bounded initial part of each series, preserve independence.*

Proof. On each finite subfamily the substitutions $X_A \mapsto u_A X_A + b_A$ define an invertible polynomial change of variables over the base field. They therefore send nonzero polynomials to nonzero polynomials. A deleted bounded initial part belongs to $\mathcal{B}_G(K)$. $\square$

Corollary 5.6 (Independent families in every valuation ball). *Every nonempty ball*

$$U(f, \delta) = \{z \in K((t^G)) : v(z - f) > \delta\}$$

contains 2^κ elements algebraically independent over $\mathcal{F}_G(K)$. It also contains an element of $\mathcal{B}_G(K)$.

Proof. Put $b = f_{\leq \delta} \in \mathcal{B}_G(K)$. Choose $\beta < \kappa$ with $a_\beta > \delta$, and define

$$Z_A = b + \sum_{\alpha \geq \beta} c_A(\alpha) t^{a_\alpha}.$$

Then $v(Z_A - b) > \delta$ and $v(b - f) > \delta$, with the usual convention for zero, so every Z_A lies in the ball. Each Z_A differs from Y_A by a bounded-support series: the omitted initial part is bounded above by a_β , and b is bounded. Corollary 5.5 proves their independence. The element b itself belongs to the same ball. $\square$

Remark 5.7 (A precise limitation of bounded truncation data). No single bounded truncation can determine algebraicity over $\mathcal{F}_G(K)$. The same truncation is shared by a base-field element and by every member of a family as in Corollary 5.6. This is an information-theoretic statement about those data, not a claim that every representation of Hahn series has an undecidable algebraicity problem.

5.3 Exact transcendence degrees

Corollary 5.8 (Cardinality-matched groups). *Suppose $\kappa = \text{cf}(G) = |G|$ and $|K| \leq 2^\kappa$, with K of characteristic zero. Then*

$$\text{trdeg}_{\mathcal{F}_G(K)} K((t^G)) = 2^\kappa.$$

Proof. The lower bound is Theorem 5.3. A Hahn series is a function from G to K , so

$$|K((t^G))| \leq |K|^{|G|} \leq (2^\kappa)^\kappa = 2^\kappa.$$

The cardinality of a field bounds its transcendence degree, proving the upper bound independently of the construction. $\square$

Corollary 5.9 (Every nonzero real-exponent subgroup). *Let $0 \neq G \subseteq \mathbb{R}$ and let K have characteristic zero and $|K| \leq \mathfrak{c} = 2^{\aleph_0}$. Then*

$$\text{trdeg}_{\mathcal{F}_G(K)} K((t^G)) = \mathfrak{c}. \quad (16)$$

In particular this holds for $K = \mathbb{Q}, \mathbb{R}, \mathbb{C}$ and $G = \mathbb{Z}, \mathbb{Q}, \mathbb{R}$.

Proof. Here $\text{cf}(G) = \aleph_0$, so the lower bound is $\mathfrak{c}$. Every well-ordered subset $S \subseteq \mathbb{R}$ is countable. Indeed, for each nonmaximal $s \in S$ there is a successor s^+ in S ; choose a rational in (s, s^+) . These intervals are disjoint, producing an injection of S minus a possible maximum into $\mathbb{Q}$. There are at most $\mathfrak{c}^{\aleph_0} = \mathfrak{c}$ countable supports in G , and at most $|K|^{\aleph_0} \leq \mathfrak{c}$ coefficient assignments to any one support. Thus $|K((t^G))| \leq \mathfrak{c}$. $\square$

Remark 5.10 (Why the full strength is not a counting argument). For $G = \mathbb{R}$ and $K = \mathbb{R}$ or $\mathbb{C}$, the base $\mathcal{F}_G(K)$ already has cardinality $\mathfrak{c}$, since it contains K and lies in a field of cardinality $\mathfrak{c}$. Equality of these cardinalities gives no lower bound at all on relative transcendence degree. The separation and coding proof is what supplies it. Nor is an independent family of maximum possible cardinality automatically a transcendence basis: maximal cardinality need not mean maximality under inclusion.

6 Explicit countable witnesses and examples

The coefficient code gives a family of optimal cardinality with integer coefficients, but its indexing uses a set-theoretic bijection. It is useful to have additional closed-form witnesses.

6.1 Factorial gaps with exponential coefficients

Theorem 6.1 (Explicit multiplicative-parameter family). *Let u be an order unit of G and let L contain $\mathbb{R}$. If $q_1, \dots, q_m > 0$ are multiplicatively independent real numbers, then*

$$Z_{q_j} = \sum_{n \geq 2} q_j^n t^{n!u} \quad (1 \leq j \leq m) \quad (17)$$

are algebraically independent over $\mathcal{F}_G(L)$. Consequently the series Z_p , indexed by the ordinary prime numbers p , are jointly algebraically independent, with positive integer coefficients.

Proof. Multiplicative independence means that $\prod_j q_j^{m_j} = 1$ with $m_j \in \mathbb{Z}$ forces every $m_j = 0$. Let $P \in \mathcal{B}_G(L)[X]$ be a nonzero polynomial of positive total degree d . At the vector $c(n) = (q_1^n, \dots, q_m^n)$ its top part is

$$P_d(c(n)) = \sum_{|v|=d} p_v Q_v^n, \quad Q_v = \prod_j q_j^{v_j}.$$

The Q_v are distinct nonzero elements of L . Enumerate the $r \geq 1$ nonzero top coefficients as $p_1, \dots, p_r$ and their associated values as $Q_1, \dots, Q_r$. For any integer $N \geq 2$, the matrix $(Q_i^{N+k})_{0 \leq k < r, 1 \leq i \leq r}$ has determinant

$$\left(\prod_{i=1}^r Q_i^N \right) \prod_{1 \leq i < j \leq r} (Q_j - Q_i) \neq 0.$$

It is therefore invertible over L , and also over $L((t^G))$. Consequently the r values $P_d(c(N+k))$, $0 \leq k < r$, cannot all vanish, since that would force every $p_i = 0$. The factorial gaps satisfy (8) for all sufficiently large n . Choose N beyond that threshold and one nonzero evaluation in the displayed block. Lemma 4.4 proves that P does not vanish on the series. This finite Vandermonde argument treats the Hahn coefficients directly; it requires no limiting argument. Clear denominators for a polynomial over the fraction field. Finally, unique factorization of ordinary integers makes any finite set of distinct primes multiplicatively independent. $\square$

An alternative proof uses Corollary 3.4 to reduce to ordinary Laurent series over $\mathbb{R}(t^u)$. The direct support proof also records the exact obstruction to cancellation over bounded-support Hahn coefficients.

6.2 A continuum family with only integer coefficients

For $\kappa = \aleph_0$ one can choose an effective enumeration of all triples

$$(s, h, \rho), \quad s \subseteq \mathbb{N} \text{ finite}, \quad h : \mathcal{P}(s) \rightarrow \mathbb{N}_{>0}, \quad \rho \in \mathbb{N}.$$

Thus for each binary sequence $A \subseteq \mathbb{N}$, formula (13) gives an integer coefficient stream. Relabeling position $n-2$ by $n \geq 2$ (and writing $c_A(n)$ for the relabeled coefficient),

$$W_A = \sum_{n \geq 2} c_A(n) t^{n!u} \quad (18)$$

is a continuum-sized independent family whenever u is an order unit. Each individual stream is computable relative to the characteristic function of A : evaluating a coefficient asks only finitely many membership questions. The family is not asserted to consist of computable streams; there are only countably many computable descriptions.

More strikingly, for $u = 1$ these witnesses lie in $\mathbb{Q}[[t]]$, yet they remain independent over $\mathcal{F}_{\mathbb{R}}(\mathbb{C})$. All ordinary complex constants and all bounded real-exponent Hahn series have already been admitted in that base. Cofinal descent explains why this does not erase ordinary formal-series transcendence.

6.3 A small catalogue of contrasting behavior

The examples below keep the exponent group explicit.

Series or setting	Consequence
$\sum_{n \geq 0} t^n, G = \mathbb{R}$	Equals $(1 - t)^{-1}$ in $\mathcal{F}_{\mathbb{R}}(\mathbb{C})$, but does not have bounded support.
$\sqrt{1 + t}, G = \mathbb{R}$	Algebraic of degree two over $\mathcal{F}_{\mathbb{R}}(\mathbb{C})$; not a bounded-support fraction.
$\sum_{n \geq 2} t^{n!}, G = \mathbb{R}$	Transcendental over $\mathcal{F}_{\mathbb{R}}(\mathbb{C})$ by Corollary 4.6.
$\sum_{n \geq 2} p^n t^{n!}, p \text{ prime}$	A fully explicit countable independent family over the same base.
$\mathbb{R} \hookrightarrow \mathbb{R}\omega + \mathbb{R} \subset \mathbf{No}$	Every series with real exponents becomes bounded in the larger exponent group.
$\text{cf}(G) > \aleph_0$	Every countable-support Hahn series already belongs to $\mathcal{B}_G(K)$.

The first three rows show why “unbounded support” is not synonymous with “transcendental.” The final two show why a transcendence assertion must name its ambient exponent group and its base field.

7 Rank structure and the unit group

This section concerns the coefficient ring itself. The field criterion is proved from the Hahn field construction. The unit classification additionally uses the known rank-one upper-support theorem, which is isolated explicitly.

7.1 An exact field criterion

For a positive element $a \in G$, define its principal convex subgroup

$$C(a) = \{g \in G : |g| \leq na \text{ for some } n \geq 1\}.$$

It is proper precisely when a is not an order unit.

Theorem 7.1 (Absence of a highest scale). *For any field K and nonzero ordered abelian group G , the following are equivalent:*

1. G has no order unit;
2. $\mathcal{B}_G(K) = \bigcup_{C \subseteq G \text{ convex}} K((t^C))$;
3. $\mathcal{B}_G(K)$ is a field.

Proof. Assume there is no order unit. If f has support in $[\ell, u]$, choose $a > 0$ with $a \geq |\ell|, |u|$. Then its support lies in the proper convex subgroup $C(a)$, so $f \in K((t^{C(a)}))$. Conversely, a proper convex subgroup C is bounded above in G : choose a positive $g \notin C$. Every positive $c \in C$ is smaller than g , since otherwise convexity would put g in C . Hence every series in $K((t^C))$ has bounded support in G .

Convex subgroups of an ordered abelian group are linearly ordered by inclusion. To see this, if neither of C, D contained the other, choose positive $c \in C \setminus D$ and $d \in D \setminus C$; whichever is smaller belongs to the other’s convex subgroup, a contradiction. Thus the union in (2) is a directed union of fields and is a field.

Finally suppose $u > 0$ is an order unit. The bounded-support element $1 - t^u$ has inverse in the Hahn field

$$(1 - t^u)^{-1} = \sum_{n \geq 0} t^{nu}.$$

This is the usual strongly summable geometric identity. Its support is unbounded because u is an order unit. The inverse is unique in the Hahn field, so no inverse can belong to $\mathcal{B}_G(K)$. Thus (3) excludes an order unit and implies (1). $\square$

Corollary 7.2 (Closed coefficient fields at unbounded rank). *Suppose G is divisible and has no order unit. If K is real closed, then $\mathcal{B}_G(K)$ is real closed. If K is algebraically closed of characteristic zero, then $\mathcal{B}_G(K)$ is algebraically closed.*

Proof. A convex subgroup of a divisible ordered abelian group is divisible: the unique n th part of an element of the subgroup has no larger absolute value, so belongs to the subgroup by convexity. The standard Hahn-field closedness criterion makes each $K((t^C))$ in Theorem 7.1 real closed or algebraically closed, respectively; this is the imported criterion recorded in [1, Fact 2.1.1].

A polynomial has only finitely many coefficients, so all of them lie in one member of the directed union. In the algebraically closed case it has a root in that member. In the real closed case the same argument supplies roots of odd-degree polynomials and square roots of positive elements, which characterize real closedness. The order throughout is the inherited Hahn order. $\square$

These fields can be proper and dense in the full Hahn field. There is no conflict with algebraic closedness: the remaining extension is transcendental. Theorem 5.3 measures how large it is.

7.2 Reduction to the highest Archimedean quotient

Suppose G has an order unit u . Define

$$C = \{g \in G : n|g| < u \text{ for every positive integer } n\}. \quad (19)$$

This is a convex subgroup. It contains zero and is closed under negation. If $g, h \in C$, applying the defining inequalities at $2n$ gives $2n(|g| + |h|) < 2u$, hence $n|g + h| < u$ for every $n \geq 1$. If $|x| \leq |g|$ with $g \in C$, the same inequalities put x in C . It is proper because $u \notin C$.

For $g \notin C$, some n satisfies $u \leq n|g|$. Since u is an order unit, $|g|$ is one too. A convex subgroup containing such a g contains every element bounded in absolute value by a multiple of $|g|$, so it must be all of G . Thus C is the unique maximal proper convex subgroup. The quotient G/C is nonzero Archimedean: otherwise a positive class dominated by all multiples below another class would generate a nonzero proper convex subgroup of the quotient, whose preimage strictly contains C .

Proposition 7.3 (Top-rank presentation). *If G is divisible and has an order unit, choose an ordered embedding of $H = G/C$ in $\mathbb{R}$ and a $\mathbb{Q}$ -linear section of the quotient map. With $M = K((t^C))$, that choice gives isomorphisms*

$$K((t^G)) \cong M((t^H)), \quad \mathcal{B}_G(K) \cong \mathcal{B}_H(M). \quad (20)$$

The full-field isomorphism is the usual iterated Hahn presentation; the second is its bounded-support restriction.

Proof. Since G and C are $\mathbb{Q}$ -vector spaces, choose a complementary subspace. Order on the quotient dominates order within C , so this identifies G with the lexicographic group $H \oplus C$. Regroup a series according to its H coordinate. Its projected support is well ordered: for any nonempty subset of that projection, take the least original supported exponent whose image lies in the subset; its image is least there. Each fiber, a translated subset of the original support, is well ordered too. Conversely, to find a least element in any nonempty subset of a lexicographic support, first choose its least outer coordinate, then its least inner coordinate in that fiber. Thus well-ordered outer support and well-ordered fibers give exactly a Hahn series on G .

Addition agrees coefficientwise. For multiplication there are finitely many outer pairs at any fixed outer exponent; within each such pair, there are finitely many inner pairs at any fixed inner exponent. These are exactly the original convolution terms. Thus the grouping is a field isomorphism, including inverses by uniqueness. This standard presentation is also recorded in [1, Fact 2.4.2].

If the original support is bounded by g , its H projection is bounded by the image of g . Conversely, if its outer support has upper bound $h \in H$, choose $h' > h$. A lift of h' bounds every fiber, regardless of its internal C support. Thus boundedness is equivalent on the two sides. $\square$

Proposition 7.4 (Imported rank-one support theorem). *If M has characteristic zero and $H \subseteq \mathbb{R}$, then for nonzero $f, g \in \mathcal{B}_H(M)$,*

$$\sup_{\mathbb{R}} \text{supp}(fg) = \sup_{\mathbb{R}} \text{supp}(f) + \sup_{\mathbb{R}} \text{supp}(g).$$

This is the monomial-localized form of [1, Proposition 3.5.1]. Choose $a, b \in H$ bounding the supports of f, g . Then $t^{-a}f, t^{-b}g$ have nonpositive support. Apply the imported one-sided theorem over M to them and translate the real suprema back by a, b ; the translations cancel to give the displayed identity. The one-sided theorem is not reproved or claimed as new here. The supremum is taken in $\mathbb{R}$ and need not be an element of H .

Corollary 7.5 (Units see exactly the highest rank). *Suppose K has characteristic zero and G is divisible with an order unit, and let C be (19). A nonzero $f \in \mathcal{B}_G(K)$ is a unit if and only if its support lies in one coset of C . Equivalently, its image in (20) is a single outer monomial with an arbitrary nonzero coefficient from $K((t^C))$. In particular,*

$$1 + t^s \in \mathcal{B}_G(K)^\times \iff g \in C.$$

Proof. For a nonzero rank-one bounded series define

$$w(f) = \sup_{\mathbb{R}} \text{supp}(f) - \min \text{supp}(f) \geq 0.$$

Proposition 7.4 and the usual valuation identity give $w(fg) = w(f) + w(g)$. If $fg = 1$, the nonnegative widths satisfy $w(f) + w(g) = w(1) = 0$, so $w(f) = 0$. Every exponent of f then lies between its minimum and that same real supremum, forcing singleton support. Conversely, if $f = at^h$ with $a \in M^\times$, its inverse $a^{-1}t^{-h}$ also has singleton support and is bounded. Transfer this statement through (20). For the last claim, if $g \neq 0$ the support is $\{0, g\}$, which lies in one coset precisely when $g \in C$. If $g = 0$, the element is 2, a nonzero scalar in characteristic zero, and the same conclusion holds. $\square$

No arbitrary-rank supremum formula has been assumed. In particular the rank-one width proof should not be repeated in G itself using a guessed Dedekind completion: at higher rank such a formula requires additional structure and need not have the same meaning.

8 Actual surreal and surcomplex realizations

8.1 The normal-form bridge and its scope

Let G be a set-sized ordered additive subgroup of $\mathbf{No}$. Conway normal form supplies the embedding

$$\iota_G : \mathbb{R}((t^G)) \longrightarrow \mathbf{No}, \quad \sum_g r_g t^g \mapsto \sum_g r_g \omega^{-g}. \quad (21)$$

The reversed sign is essential: increasing Hahn exponents correspond to decreasing Conway exponents. We use the standard normal-form isomorphism and its field compatibility, as recorded in

[1, Section 2.3] and developed in [2]. This is an imported foundational theorem, not an assertion about convergence of real-indexed sums in the fine surreal topology.

Applying (21) to real and imaginary coefficients gives a field embedding

$$\iota_G^{\mathbb{C}} : \mathbb{C}((t^G)) \longrightarrow \mathbf{No}[i].$$

Indeed, coefficientwise real and imaginary parts have well-ordered supports, their union is well ordered, and complex multiplication agrees with multiplication in the quadratic extension. We henceforth identify these Hahn fields with their concrete images, but continue to name G in every bounded-support base.

8.2 Real-exponent surreal numbers

Define the concrete subfields

$$\mathcal{H}_{\mathbb{R}} = \left\{ \sum_{g \in S} r_g \omega^{-g} : S \subseteq \mathbb{R} \text{ well ordered, } r_g \in \mathbb{R} \right\}, \quad \mathcal{H}_{\mathbb{C}} = \mathcal{H}_{\mathbb{R}}[i].$$

Let $\mathcal{B}_{\mathbb{R}}$ and $\mathcal{B}_{\mathbb{C}}$ require S to be bounded above in $\mathbb{R}$, and put $\mathcal{F}_{\mathbb{R}} = \text{Frac } \mathcal{B}_{\mathbb{R}}$ and $\mathcal{F}_{\mathbb{C}} = \text{Frac } \mathcal{B}_{\mathbb{C}}$.

Theorem 8.1 (A continuum of independent actual infinitesimals). *There is a family of positive infinitesimal surreal numbers*

$$\eta_A = \sum_{n \geq 2} c_A(n) \omega^{-n!} \quad (A \subseteq \mathbb{N}),$$

with positive integer coefficients, algebraically independent even over $\mathcal{F}_{\mathbb{C}}$. Moreover

$$\text{trdeg}_{\mathcal{F}_{\mathbb{R}}} \mathcal{H}_{\mathbb{R}} = \text{trdeg}_{\mathcal{F}_{\mathbb{C}}} \mathcal{H}_{\mathbb{C}} = \mathfrak{c}.$$

The separate explicitly given family

$$\eta_p = \sum_{n \geq 2} p^n \omega^{-n!} \quad (p \text{ an ordinary prime})$$

is also algebraically independent over $\mathcal{F}_{\mathbb{C}}$.

Proof. Use the countable coefficient code and factorial separated system in the complex-coefficient theorem. Its coefficients happen to be positive integers, so the witnesses lie in the real subfield. Their leading term is a positive real multiple of ω^{-2} ; the remaining terms have strictly smaller Conway monomials. Thus they are positive infinitesimals. The exact degrees follow from Corollary 5.9, and the prime family from Theorem 6.1, all transported by (21). $\square$

This is not an assertion of transcendence over all surreals or all surcomplex numbers: the witnesses themselves belong to those proper-class fields. It is a relative algebraic theorem over explicitly specified set-sized subfields.

8.3 Maximal families at every infinite regular cardinal

Let κ be an infinite regular cardinal, regarded as an initial ordinal, and form the ordered additive subgroup of actual surreal numbers

$$G_{\kappa} = \bigoplus_{\alpha < \kappa} \mathbb{Q} \omega^{\alpha}. \quad (22)$$

The direct sum means finite rational linear combinations. Comparison is by the largest exponent with a nonzero coefficient. This group has cardinality κ , is divisible, has cofinality κ , and has no order unit. For the cofinality assertion, any family of fewer than κ finite supports uses a bounded set of indices, by regularity; a larger basis monomial dominates it. Conversely the ω^α are cofinal. The absence of an order unit follows by moving one index past the largest index occurring in a proposed unit.

Let $\mathcal{H}_\kappa^\mathbb{R}$ and $\mathcal{H}_\kappa^\mathbb{C}$ be the real and complex Hahn fields on this concrete exponent group, interpreted in **No** and **No**[i]. Let $\mathcal{B}_\kappa^\mathbb{R}$ and $\mathcal{B}_\kappa^\mathbb{C}$ be their bounded-support rings.

Theorem 8.2 (Arbitrarily large, explicit-support surreal families). *For every infinite regular cardinal κ :*

1. $\mathcal{B}_\kappa^\mathbb{R}$ is a real closed field and $\mathcal{B}_\kappa^\mathbb{C}$ is an algebraically closed field.
2. There are 2^κ positive infinitesimal surreal numbers

$$\eta_A = \sum_{\alpha < \kappa} c_A(\alpha) \omega^{-\omega^{\alpha+1}} \quad (A \subseteq \kappa), \quad (23)$$

all with positive integer coefficients and exactly the displayed common support, algebraically independent over $\mathcal{B}_\kappa^\mathbb{C}$.

3. Both exact relative transcendence degrees are 2^κ :

$$\text{trdeg}_{\mathcal{B}_\kappa^\mathbb{R}} \mathcal{H}_\kappa^\mathbb{R} = \text{trdeg}_{\mathcal{B}_\kappa^\mathbb{C}} \mathcal{H}_\kappa^\mathbb{C} = 2^\kappa.$$

No element with a support of smaller cardinality is transcendental over its corresponding bounded-support base.

Proof. The first assertion is Corollary 7.2. For the second, the concrete choices

$$a_\alpha = \omega^{\alpha+1}, \quad b_\alpha = \omega^\alpha \quad (\alpha < \kappa)$$

form a separated exponent system in G_κ . Earlier a_β are at most b_α . For every finite n , $a_\alpha > nb_\alpha$, and the b_α are cofinal. Thus the two inequalities in (7) follow exactly as in the uncountable construction, and also hold when $\kappa = \aleph_0$. Apply Theorem 5.3 over $\mathbb{C}$ and then the Conway embedding. Since the base ring is already a field, its fraction field is itself. The leading term of each number is a positive integer multiple of $\omega^{-\omega}$, proving positivity and infinitesimality.

Finally $|G_\kappa| = \text{cf}(G_\kappa) = \kappa$, while $|\mathbb{R}| = |\mathbb{C}| = \mathfrak{c} \leq 2^\kappa$. Corollary 5.8 supplies the upper bounds as well as the equality. Optimality is Corollary 5.4. $\square$

No continuum hypothesis or generalized continuum hypothesis is used. The regularity assumption on κ in this concrete example is genuine: for a singular initial ordinal the direct sum in (22) has cofinality $\text{cf}(\kappa)$, not κ . The general theorem still applies to that group with its actual cofinality, but it does not give the displayed 2^κ equality by the same argument.

8.4 Enlarging a workspace can erase the base distinction

Consider

$$G = \mathbb{R}\omega + \mathbb{R} \subset \mathbf{No}, \quad H = \mathbb{R} \subset G.$$

Every real number is smaller than ω in this ordered group, so H is not cofinal. All real-exponent Hahn series therefore have support bounded above in G . In particular every witness of Theorem 8.1

belongs to $\mathcal{B}_G(\mathbb{C})$, even though it was transcendental over $\mathcal{F}_\mathbb{C}$ in the smaller workspace. This is the concrete collapse in (6), not a failure of field embedding.

At the full class level every set of surreal exponents is bounded above by a surreal number. Consequently the naive class-wide version of “bounded support in **No**” imposes no restriction on a Conway normal form. All the nontrivial statements in this article are deliberately workspace-relative. A set-sized cofinal support in G_κ is not a cofinal subclass of all of **No**.

The valuation topology used for density is likewise the topology of the named Hahn workspace. For a constructed family member, the bounded initial Hahn truncations

$$Y_{A,<\beta} = \sum_{\alpha<\beta} c_A(\alpha)t^{a_\alpha} \quad (\beta < \kappa)$$

form a convergent net there: each belongs to $\mathcal{B}_G(K)$, and $v(Y_A - Y_{A,<\beta}) = a_\beta$ is cofinal in G . For $\kappa = \aleph_0$ these are finite partial sums. For uncountable κ they are generally infinite bounded-support series. Indeed the net of all finite subsums does not converge to Y_A then: every finite subset omits some index $n < \omega$, leaving a nonzero coefficient below a_ω , so no such subsum belongs to $\{z : v(z - Y_A) > a_\omega\}$. It cannot converge to any other Hahn series either, since convergence would force every coefficient to equal that of Y_A .

Even the convergent initial-truncation net does not converge in the full surreal fine topology. Choose a surreal exponent θ above the set-sized group G . Every nonzero truncation error has a leading Hahn exponent in G and magnitude greater than $\omega^{-\theta}$, so the fine neighborhood of that radius is never reached. This retains the workspace distinction in the repository’s foundations [4] and its notation guide, under “Topologies and strong summation” [8].

9 Relation to other reports in this collection

This section was added at editorial assembly in commit ed88b8f and its statement comparisons were revisited at b4275b5. It compares statements only; no theorem of another report is used in any proof above, and no theorem here is claimed to follow from, or to imply, one of theirs unless this section says so and gives the reason.

9.1 Tail-spans and differential transcendence

The tail-span report [5] is another source of algebraic independence in Hahn fields in the collection. The two reports answer different base-field problems.

Base fields. Its base is a *full* Hahn field $M((t^\Gamma))$ over a smaller characteristic-zero coefficient field M , and its transcendence comes from the coefficients: by its coefficient-field criterion (`tail:thm:coefficient`) a series with coefficients in an algebraic closure of M is algebraic over $M((t^\Gamma))$ exactly when its coefficients generate a finite extension of M . Here the base $\mathcal{F}_G(K)$ keeps the coefficient field of the witnesses (and in Theorem 3.2 may even enlarge it), and transcendence comes from the support. The actual numbers make the difference concrete. With $t = \omega^{-1}$, the witnesses $\eta_A = \sum_{n \geq 2} c_A(n)t^{n!}$ and η_p of Theorem 8.1 have integer coefficients and integer exponents, so they are *elements* of the tail-span base $B_0 = \mathbb{Q}((t^\mathbb{Q}))$. Conversely $\sqrt{2}$, or the bounded real-coefficient series $\sum_{n \geq 2} \sqrt{p_n} t^{1-1/n}$, lies in $\mathcal{F}_\mathbb{R}(\mathbb{R})$ but not in B_0 . Neither base contains the other, and neither report’s independence theorem implies the other’s. At uncountable cofinality the two criteria separate further: if $\text{cf}(G) > \aleph_0$ and (γ_n) is a strictly increasing sequence in G , the series $\sum_n \sqrt{p_n} t^{\gamma_n}$ has countable support, so it lies in $\mathcal{B}_G(\overline{\mathbb{Q}} \cap \mathbb{R})$ by Lemma 2.4, while its coefficients generate an infinite extension of $\mathbb{Q}$, so it is transcendental over $\mathbb{Q}((t^G))$ by `tail:thm:coefficient`.

What overlaps. Both reports prove a relative transcendence degree $2^{\aleph_0}$ for set-sized Hahn fields with a countably cofinal exponent group, with explicit positive infinitesimal surreal witnesses, and both obtain the upper bound by the same separate cardinality count rather than from the witnesses: the tail-span report for $\text{trdeg}_{\mathbb{Q}((t^{\mathbb{Q}}))}(\overline{\mathbb{Q}} \cap \mathbb{R})((t^{\mathbb{Q}}))$ (tail:thm:surreal), this report for $\text{trdeg}_{\mathcal{F}_G(K)} K((t^G))$ with $0 \neq G \subseteq \mathbb{R}$ and $|K| \leq \mathfrak{c}$ (Corollary 5.9, Theorem 8.1). Both state that an independent family of maximal cardinality is not thereby a transcendence basis, and both are relative to named set-sized workspaces. Both show that enlarging the exponent group along a noncofinal embedding changes the answer for the same surreal expansion: its tail:thm:closure example embeds $\mathbb{Q}$ in lexicographic $\mathbb{Q}^2$ by $q \mapsto (0, q)$, whose image is bounded above by $(1, 0)$, and loses approximability by algebraic elements; the example $\mathbb{R} \subset \mathbb{R}\omega + \mathbb{R}$ of Section 8 makes every old series a bounded-support coefficient and so erases transcendence over the bounded base. The phenomena differ; the common cause is the noncofinal embedding. Finally, both warn that finite or bounded prefix data do not decide algebraicity; the tail-span report proves an undecidability result for computable coefficient streams (tail:prop:undecidable), while this report proves no undecidability statement (the remark after Corollary 5.6).

What differs. The tail-span report proves differential independence for the Berarducci–Mantova derivation and for surcomplex analytic functions; this report contains no differential or analytic statement. It classifies all relations among its sums through the cofinite span (tail:thm:tail); this report does not classify relations, but its descent theorem transports relation ideals between bases, which has no counterpart there. Its displayed independent family uses almost-disjoint infinite index sets. This is a sufficient construction, not a necessary condition for independence. The unrestricted family $(\xi_A)_{A \subseteq \mathbb{N}_{>0}}$ satisfies the linear relation $\xi_{A \cup B} + \xi_{A \cap B} = \xi_A + \xi_B$ (tail:rem:indexfamily). The family (η_A) here is indexed by all subsets and is independent, because c_A is the finite-pattern code of Lemma 5.1, not an indicator function; this is why the witnesses were renamed (Section 1.4). Its displayed actual-surreal differential family uses $\Gamma = \mathbb{Q}$; its coefficient-field and cofinite-span theorems allow arbitrary set-sized ordered value groups. This report proves that $\text{cf}(G)$ is the exact support threshold over the bounded base.

9.2 Entire functions at arbitrary rank

The entire-functions report [6] uses the same invariant, $\text{cf}(G)$, and the same order unit (Section 2.3), for a different object: the ring of strongly entire power series in one variable over $\mathbb{C}((t^\Gamma))$, with Γ divisible in its main theorems. The parallels are these.

- *The same threshold $\aleph_0$.* There, nonpolynomial entire series exist exactly when $\text{cf}(\Gamma) = \aleph_0$ (ent:cor:cofinality). Here, a transcendental element with countable support exists exactly when $\text{cf}(G) = \aleph_0$ (Corollary 5.4). Both rest on the fact that a countable subset of a group of uncountable cofinality is bounded above.
- *The order unit.* There, the order unit separates the second and third cases of the cofinality–rank trichotomy (ent:thm:main) and decides which restricted series are units (ent:thm:units). Here, it decides whether $\mathcal{B}_G(K)$ is a field (Theorem 7.1); the obstruction is the geometric series $(1 - t^u)^{-1} = \sum_n t^{nu}$, whose support is unbounded exactly because u is an order unit.
- *Cofinal and noncofinal enlargement.* There, a nonpolynomial entire series stays entire over a larger group exactly when the old group is cofinal in it (ent:thm:main-extension(b)). Here, the descent map (4) is injective exactly when the subgroup is cofinal (Theorem 1.2). These are different statements about different objects, proved independently; this report assumes no divisibility for its descent theorem.

- *GCD rings*. The entire ring there is proved to be a GCD domain. Appendix A concerns the bounded-support Hahn ring, a different ring, and remains conditional on $(\text{GCD}_{\mathbb{R}, \overline{\mathbb{K}}})$; nothing in the entire-functions report supplies that hypothesis.

9.3 Hidden negative directions

The report on hidden negative directions in surcomplex linear algebra [7], placed in the same commit as this one, proves a third kind of independence. Its base $\mathcal{P}_\Gamma$ is the union of the fields $\mathbb{C}((t^H))$ over finitely generated subgroups $H \subseteq \Gamma$ (the Puiseux field when $\Gamma = \mathbb{Q}$). Its binary-prefix almost-disjoint family of infinite prime sets gives tails $h_A = \sum_{p \in A} t^{(1-1/p)\delta}$ independent over that field by fresh prime denominators. This does not assert independence for all infinite prime sets.

For $\Gamma = \mathbb{Q}$, the fields $\mathcal{P}_\mathbb{Q}$ and $\mathcal{F}_\mathbb{Q}(\mathbb{C})$ are incomparable. Each selected h_A has support bounded above by δ , so it belongs to $\mathcal{B}_\mathbb{Q}(\mathbb{C}) \subseteq \mathcal{F}_\mathbb{Q}(\mathbb{C})$, but it lies outside $\mathcal{P}_\mathbb{Q}$. Conversely, the witnesses W_A of (18) with $u = 1$ belong to $\mathbb{C}((t^\mathbb{Z})) \subseteq \mathcal{P}_\mathbb{Q}$ and are transcendental over $\mathcal{F}_\mathbb{Q}(\mathbb{C})$. For groups without an order unit the containment behaves differently: every finitely generated subgroup is bounded above, so $\mathcal{P}_\Gamma \subseteq \mathcal{B}_\Gamma(\mathbb{C})$. Neither the ring nor its fraction field should be confused with a full Hahn coefficient field.

9.4 Foundations and notation

The distinction between the intrinsic valuation topology of a named Hahn workspace and the fine topology of **No**, used in Section 8, is the collection's standing convention [8]. The Conway normal-form embedding (21) is imported as a standard theorem here; its formal counterpart in the collection is tracked in docs/NORMAL_FORM_BRIDGE.md [9].

9.5 Independent surreal copies

This subsection was added in batch 31. The independent-copies report [11] compares its bounded-support section with this report. Its identification of $\mathcal{F}_H(k)$ with the fraction field of the omnific-type ring of series with negative support plus a constant in $\mathbb{Z}$ or $\mathbb{Z}[i]$, for $k = \mathbb{R}, \mathbb{C}$ (`isc:prop:boundedfrac`) is the localization of Proposition 2.2, with the constant coefficient restricted. Its one-series theorem `isc:thm:cofinalgap` is compared with Corollary 4.6 and Theorem 5.3 after that corollary. Its coset slices are the projections π_r of Lemma 3.1. With them it proves a statement not made here (`isc:thm:sliceddisjoint`): the full Hahn fields on two subgroups are linearly disjoint over the full Hahn field on their intersection, with no cofinality hypothesis. Here the same projections serve descent over a cofinal subgroup (Theorem 3.2). No statement of this report changes.

10 Verification, formalization, and remaining questions

10.1 What was checked computationally

The accompanying code/`verify.py` uses exact rational arithmetic and sparse finite series. It checks the finite convolution identities underlying coset projection, coefficient-functional descent, the auxiliary-polynomial expansion, support-band separation, and the tail bound. It also checks finite-pattern coding, integer-grid detection, factorial-gap inequalities, and explicit multiplicative-parameter examples. Its seed, counts, and outputs are recorded in `data/verification.json`. The delivered run passed 3,163 exact assertions in twelve groups. This is an assertion count, not a count of independently proved theorems. When the report was placed in the collection, the unchanged script was rerun on a copy under Python 3.14.4 and reproduced the delivered record in every field except

the recorded Python version (the delivered run used 3.13.5). The countable coefficient code itself is `code/coefficients.py`, which `verify.py` imports.

These checks are regression tests of the finite mechanisms. They are not proofs of the arbitrary-support, transfinite, or cardinal assertions. In particular, a finite enumeration does not verify that a coding fiber has cardinality κ , and a finite truncation does not verify algebraic independence of an infinite family. Those assertions are established by the proofs above. The script uses only the Python standard library and can be run from the report directory with

```
python code/verify.py -output <scratch>/verification-local.json.
```

Its default output is a file `verification.json` in the current directory; an explicit scratch path keeps the delivered record in `data/` unchanged. The delivered verification record is separate from the PDF build record `data/build_audit.json`, which describes the delivered 24-page PDF, not the PDF of this report.

10.2 A feasible formalization decomposition

No Lean proof is supplied or claimed. The results nevertheless break into clear interfaces, without a need to formalize all surreal numbers first. The first interface is Hahn subseries extraction and its module identity (2), followed by coefficient-linear maps (3). The second is a bounded-support subring with finite support bounds and an explicit fraction-field embedding. The third is a finite multivariable polynomial expansion lemma giving (10)–(12). These are the main algebraic proof obligations.

Only after those interfaces are complete is it necessary to add the cofinality construction, finite-pattern coding, and cardinal estimates. The actual-surreal consequences are then transported through the Conway-normal-form embedding. Such a transport must be linked to the repository’s exact proved normal-form bridge, not merely to an abstract Hahn field with the same informal description. The current state of that bridge in the Lean development is recorded in `docs/NORMAL_FORM_BRIDGE.md` [9]; no statement of this report has an implementation mapping in the collection’s formalization ledger. Its 30 standard result environments are listed in the statement inventory there, which records source coverage rather than Lean proofs. The optional GCD descent in Appendix A belongs to a separate dependency branch.

10.3 Questions left by these theorems

The results do not determine $\text{trdeg}_{\mathcal{F}_G(K)} K((t^G))$ for every large ordered group: the lower bound $2^{\text{cf}(G)}$ can be smaller than the cardinality of the Hahn field, and no general matching upper bound is asserted. For coefficient fields of finite characteristic, the descent and field criterion still hold, and the independent-family argument works with a countably infinite alphabet in an infinite coefficient field. The positive-integer formulation is specific to characteristic zero. Finite coefficient fields require a different coding argument and are not covered by the independence theorem here.

A second question is whether stronger naturality can be imposed on the 2^κ -sized independent family. The existence proof chooses an enumeration of finite tables. It is uniform once that enumeration is fixed, but it is not invariant under every automorphism of the exponent group. The explicit prime family removes that choice at the cost of being only countable.

Finally, the GCD question below concerns divisibility inside the bounded ring, while the main theorem concerns algebraic independence over its fraction field. Neither assertion substitutes for the other. In particular, proving a ring to be GCD would not make the huge transcendental extension disappear.

A Normalized GCD descent and a published question

This appendix is logically independent of the transcendence theorems. It provides a full proof of a conditional reduction rather than importing a recent claimed result as an established premise.

A.1 The input and the conclusion

For a characteristic-zero field K , let $\bar{K}$ be an algebraic closure. Consider the hypothesis

$$\mathcal{B}_{\mathbb{R}}(\bar{K}) \text{ is a GCD domain.} \quad (\text{GCD}_{\mathbb{R}, \bar{K}})$$

A GCD domain is an integral domain in which every pair has a greatest common divisor, understood up to units. The existence of a GCD for a finite nonempty family follows by induction. By Proposition 7.4, the units of the rank-one bounded-support ring are exactly the nonzero monomials.

For a nonzero bounded series define

$$N(f) = \text{lc}(f)^{-1} t^{-v(f)} f. \quad (24)$$

It has least exponent zero and coefficient 1 there. Two nonzero bounded series are associates precisely when their normalizations agree.

Theorem A.1 (Normalized GCD descent). *Assume $(\text{GCD}_{\mathbb{R}, \bar{K}})$. For every nonzero ordered subgroup $H \subseteq \mathbb{R}$, $\mathcal{B}_H(K)$ is a GCD domain. More precisely, the normalized GCD in $\mathcal{B}_{\mathbb{R}}(\bar{K})$ of any finite family from $\mathcal{B}_H(K)$, not all zero, belongs to $\mathcal{B}_H(K)$ and is a GCD there. No divisibility or completeness assumption on H is required.*

A.2 Characters separate exponent cosets

Lemma A.2. *Let A be an abelian group and L an algebraically closed field of characteristic zero. For every $0 \neq a \in A$, there is a character $\chi : A \rightarrow L^\times$ with $\chi(a) \neq 1$.*

Proof. On the cyclic group generated by a , choose a primitive m th root of unity as the image of a if it has finite order $m > 1$, and choose 2 if it has infinite order. In either case the resulting character separates a from zero.

We explain why this character extends. The abelian group $L^\times$ is divisible, because every nonzero scalar has an n th root. Suppose a character has been defined on a subgroup $D \subseteq A$, and choose $x \notin D$. If no nonzero integer multiple of x lies in D , give x any nonzero image. Otherwise the integers n with $nx \in D$ form $m\mathbb{Z}$ for a least positive m . Choose $y \in L^\times$ with $y^m = \chi(mx)$, and set

$$\tilde{\chi}(d + nx) = \chi(d)y^n.$$

If $d + nx = d' + n'x$, then $(n - n')x = d' - d \in D$. In the nontrivial case $n - n' = km$, so $\chi(d') = \chi(d)\chi(mx)^k = \chi(d)y^{n-n'}$ and the assigned values agree. In the other case the representation is unique. The formula is multiplicative with respect to addition, so it is an extension. Order the extensions by domain inclusion. Their domains are subgroups of the set A , and the union along a chain is again a character extending the given one. Zorn's lemma gives a maximal extension; the preceding one-element construction forces its domain to be all of A . $\square$

Proof of Theorem A.1. Let d be the normalized ambient GCD of $f_1, \dots, f_m \in \mathcal{B}_H(K)$. For every character $\chi : \mathbb{R}/H \rightarrow \bar{K}^\times$, define

$$T_\chi \left(\sum_r a_r t^r \right) = \sum_r a_r \chi(r + H) t^r.$$

This is an automorphism of the Hahn field and of its bounded-support subring: it preserves support and its multiplicativity follows at each finite convolution coefficient. Its inverse uses χ^{-1} . It takes normalized series to normalized series, since the coefficient at exponent zero remains 1. It fixes every f_j . Consequently $T_\chi(d)$ is another normalized GCD of the same family. Uniqueness of normalization implies $T_\chi(d) = d$.

If an exponent $r \notin H$ occurred in d , Lemma A.2 would supply a character with $\chi(r + H) \neq 1$. Its nonzero coefficient would then change, contradicting the equality. Hence $\text{supp}(d) \subseteq H$.

Next every automorphism in $\text{Aut}(\overline{K}/K)$ acts coefficientwise and again preserves normalization and fixes the input family. Thus it fixes every coefficient of d . In characteristic zero the fixed field of all these automorphisms is K . Indeed, an algebraic element outside K has a separable minimal polynomial with another conjugate; an embedding taking it to that conjugate extends to an automorphism of the algebraic closure. Therefore every coefficient of d lies in K . Its support is bounded in $\mathbb{R}$ and lies in the nonzero subgroup H , which is cofinal in $\mathbb{R}$; it is bounded in H . Thus $d \in \mathcal{B}_H(K)$.

Each quotient f_j/d belongs to $K((t^H))$ because this is a field. It is bounded in the ambient ring by divisibility of the ambient GCD, so cofinality makes it bounded in H . Hence d divides every f_j in the base ring. If $e \in \mathcal{B}_H(K)$ is any common divisor, then $e \neq 0$ because at least one input is nonzero. Ambient universality gives $e \mid d$ there. The quotient d/e is again in $K((t^H))$ and ambient bounded, hence in $\mathcal{B}_H(K)$. This proves base-ring universality. The all-zero family can separately be assigned GCD zero. $\square$

A.3 Relation to the pre-Schreier question

L’Innocente and Mantova ask, immediately after their Example 9.1.3, whether $\mathcal{B}_H(K)$ is pre-Schreier for a divisible Archimedean group H [1]. Since a GCD domain is pre-Schreier, Theorem A.1 gives a positive answer *conditional on* $(\text{GCD}_{\mathbb{R}, \overline{K}})$. The latter hypothesis also follows if the one-sided ring $\overline{K}((t^{\mathbb{R} \leq 0}))$ is a GCD domain, since Proposition 2.2 identifies its monomial localization with $\mathcal{B}_{\mathbb{R}}(\overline{K})$, and localization preserves the GCD property. These two general commutative-algebra implications are standard; the new argument in this appendix is the normalized character descent.

There is a further conditional consequence at arbitrary divisible rank. Suppose the real-exponent GCD input holds over the algebraic closure of *every* characteristic-zero coefficient field. If G has no order unit, its bounded-support ring is already a field. If it has an order unit, Proposition 7.3 writes its bounded ring as $\mathcal{B}_H(M)$ with $H \subseteq \mathbb{R}$ and $M = K((t^{\mathbb{C}}))$. Applying the descent theorem to this coefficient field proves that $\mathcal{B}_G(K)$ is GCD. The required ambient hypothesis is now about $\mathcal{B}_{\mathbb{R}}(\overline{M})$, where $M = K((t^{\mathbb{C}}))$; an input only for $\overline{K}$ would not suffice. Thus a universal rank-one input would settle bounded-support GCD arithmetic at every divisible rank.

Warning A.3 (Recent external claim, not a premise of the main results). The source audit records that Dan Abramov’s public 2026 project `gaearon/conway-refinement` advertised a Lean proof that the one-sided real-exponent Hahn ring is GCD [10]. Its standalone statement `HahnSeriesGCD.lean` has a characteristic-zero hypothesis and the relevant nonpositive-support definition. The project described its work as provisional at that audit. The source audit read the statement and README but did not build or verify the full proof dependency chain or independently validate its semantics. Accordingly this article retains $(\text{GCD}_{\mathbb{R}, \overline{K}})$ as an explicit hypothesis. It does not claim an independent resolution of Conway’s refinement conjecture or an unconditional resolution of the published bounded-support pre-Schreier question.

B Detailed audit and dependency ledger

B.1 Novelty candidates versus imported facts

Statement	Status in this draft	Dependency boundary
Cofinal coefficient–exponent descent	Proposed contribution; full proof in Theorem 3.2.	Coset extraction, a coefficient-linear functional, finite convolution.
Exact cofinality converse	Proposed companion; Sections 3 and 5.	Noncofinal supports become bounded; properness follows from independence.
$2^{\text{cf}(G)}$ independent family on optimal support	Principal proposed contribution; Theorem 5.3.	Full support-gap and coding proofs; no GCD input.
Earlier nonrationality/cofinality obstruction	Prior result, not claimed new.	L’Innocente–Mantova, Proposition 2.4.5.
Finite-pattern coding and integer-grid detection	Elementary background mechanisms, proved here.	Choice and finite polynomial root bounds.
Real-exponent degree $\mathfrak{c}$ and arbitrary-regular-cardinal surreal examples	Consequences of the principal theorem, with explicit constructions.	Independent cardinal upper bounds; standard normal form for actual surreals.
Field criterion and top-rank unit formulation	Structural companion results; modest novelty claim.	Hahn fields; top-rank grouping is standard; rank-one upper-support theorem imported.
Hahn real/algebraic closedness and Conway normal form	Imported foundations, not new proofs.	Standard results explicitly identified in the text.
Normalized GCD descent	Proposed conditional reduction; full implication proof.	Existence of the ambient GCD remains a hypothesis.
Lean certification of this article	Not claimed.	Finite exact tests are not a proof-assistant check.
Comparison with other reports (Section 9)	Editorial addition at placement; no new theorem.	Statements of the cited reports, quoted by label; none is used in the proof of a theorem here.

B.2 Repository comparison actually performed

The repository’s root overview, documentation catalogue, current docs/new tree, and the tail-span report’s detailed README were read through the GitHub connector. The initial inspected revision was `048b72cf7cbfc8ab246e4f73788c10460cb3f6e0`. The docs/new tree at that revision contained its README rather than an additional unreviewed article collection. Targeted repository searches for `lacunary` and `bounded support` returned no indexed matches; that negative search result alone is not proof of absence. The comparison relies principally on the catalogue and the closest report’s explicitly described base fields and conclusions.

The audit was not a line-by-line comparison against all 36 full manuscripts, all retired sources in Git history, or all Lean declarations. It therefore supports the formulation “not identified in the reviewed

material,” not a guaranteed absence from every historical file. The delivered `research_audit.md` records this limitation alongside the literature queries and the exact mathematical strengthening over the closest published cofinality result.

B.3 Repository check made when the report was placed

The placement and assembly history is now checked against the retained commits `30dfb4f` and `ed88b8f`. Revision `048b72cf7cbfc8ab246e4f73788c10460cb3f6e0` exists; its catalogue lists 36 reports, and the later revision `6e34651fe640` is its successor, concerning Hahn–Hilbert restrictions and closed range. At the pin, a case-insensitive literal search in the tracked documentation, excluding `docs/new`, finds no lacunary. The phrase “bounded support” has two textual matches, in the analytic-geometry guide and article, describing the same unrelated “bounded support cuts” topic in a fibre of maximal ideals.

At assembly, lacunary does occur in this newly added report and its delivered audit, so a claim that it occurs nowhere in that tree would be false. The same search outside this report still finds none at that specific commit. Separately, the three-duals report’s `duals:ex:bounded-support` discusses a vector-valued Hahn series with support bounded above. That is a conceptual comparison, not a literal hit for the phrase “bounded support” at assembly. It concerns approximation in Hahn vector spaces, not the ring (1) or its fraction field. These dated searches make no absence claim about the current or future collection; the mathematical comparisons are in Section 9.

B.4 Proof audit: the points most vulnerable to mistakes

The proofs depend on several distinctions that deserve an explicit final check. The base is the fraction field of bounded-support series, so every putative relation must first have its denominators cleared. At limit indices the earlier partial series are not finite; the proof uses bounded support, not finite support. A top homogeneous polynomial need not be nonzero on an arbitrary coefficient sequence, which is why the coding lemma realizes a complete positive-integer grid cofinally. The coding family has size 2^κ but uses only κ positions; finite patterns, not arbitrary infinite patterns, are what it realizes.

Coset projection is not a ring homomorphism; it is linear over the Hahn field on the subgroup. Likewise the coefficient functional is not a field homomorphism; it is linear over the smaller coefficient field. Cofinality is exactly what makes their output coefficients bounded in the smaller group. For units, upper-support multiplicativity is used only after passing to an Archimedean quotient. For concrete surreal examples, the monomial map is Conway’s $x \mapsto \omega^x$, and the Hahn valuation sign is reversed in (21). Finally, a maximal-cardinality independent family is not claimed to be a transcendence basis.

B.5 What the deliverable establishes

The unconditional core is a collection of relative algebraic theorems with detailed proofs and explicit witnesses, including the sharp support threshold, coefficient–exponent descent, and arbitrarily large actual surreal examples. Their mathematical validity can be assessed directly from the given arguments and the named standard foundations. Their publication novelty requires further expert literature review. The optional GCD reduction has an additional stated hypothesis and should not be promoted into an unconditional open-problem solution by silently accepting an unverified external claim.

References

- [1] Sonia L’Innocente and Vincenzo Mantova, *A factorisation theory for generalised power series and omnific integers*, *Advances in Mathematics* **442** (2024), article 109513.
doi:10.1016/j.aim.2024.109513. Version examined: [arXiv:1710.07304v5](#), 22 January 2024.
Relevant locations: Fact 2.1.1, Section 2.3, Fact 2.4.2, Proposition 2.4.5, Proposition 3.5.1, and the question after Example 9.1.3.
- [2] Harry Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series 110, Cambridge University Press, 1986. doi:10.1017/CBO9780511629143.
Cited for standard normal-form foundations; bibliographic data also recorded in [1].
- [3] The Stacks Project Authors, *The Stacks Project*, Section 9.27, “Linearly disjoint extensions,” Tag 09IC. Accessed 22 September 2026.
- [4] Vladimir Reshetnikov, *Surreal*, repository documentation catalogue, docs/README.md, revision 048b72cf7cbfc8ab246e4f73788c10460cb3f6e0. [Versioned repository source](#). Accessed 22 September 2026.
- [5] *Tail-spans and differential transcendence in surreal and surcomplex Hahn fields*, AI-assisted research report in VladimirReshetnikov/Surreal, docs/surreal/tail-spans-and-differential-transcendence/. Detailed README examined at the same revision as [4]. [Versioned report overview](#). Not treated as a refereed source. Labels quoted in Section 9 were checked against its article.tex at assembly commit ed88b8f; the current comparison was revisited as stated in Section 9.
- [6] *Cofinality, Factorization and Scalar Extension for Entire Hahn Functions at Arbitrary Rank*, AI-assisted research report in VladimirReshetnikov/Surreal, docs/surcomplex/entire-functions-at-arbitrary-rank/. Labels quoted in Section 9 were checked at assembly commit ed88b8f and revisited as stated there. Not treated as a refereed source.
- [7] *Hidden Negative Directions in Surcomplex Linear Algebra*, AI-assisted research report in VladimirReshetnikov/Surreal, docs/surcomplex/hidden-negative-hermitian-directions/, placed in commit 30dfb4f and assembled in ed88b8f. Cited here by theorem title (“Independent prime-tail family”) and by the definition of its base field $\mathcal{P}_T$. Not treated as a refereed source.
- [8] Notation guide of VladimirReshetnikov/Surreal, docs/NOTATION.md, section “Topologies and strong summation” and the table “Scalars, workspaces, and size”, at assembly commit ed88b8f.
- [9] Implementation note docs/NORMAL_FORM_BRIDGE.md of VladimirReshetnikov/Surreal, on the Lean normal-form bridge, at assembly commit ed88b8f.
- [10] Dan Abramov, *A Proof of Conway’s Refinement Conjecture*, public provisional project gaearon/conway-refinement, 2026. [Project README](#) and [standalone Hahn-series GCD statement](#). Accessed 22 September 2026. The proof build was not independently rerun in this audit; the stated GCD input is retained as a hypothesis.
- [11] *Independent Surreal Copies and Arithmetic Intersections*, AI-assisted research report in VladimirReshetnikov/Surreal, docs/surreal/independent-surreal-copies/, written in

commit 781b19e (batch 31). Labels quoted after Corollary 4.6 and in Section 9.5 were checked against its `article.tex` at that commit. Not treated as a refereed source.